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The Immunomodulating Properties of Tiger Milk Medicinal Mushroom, *Lignosus rhinocerus* TM02® Cultivar (Agaricomycetes) and Its Associated Carbohydrate Composition

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ABSTRACT: Natural compounds found in *Lignosus rhinocerus* like polysaccharides and polysaccharide–protein complexes have the capabilities to modulate the immune system. It possesses antitumor and anti-inflammatory properties and is commonly used in Southeast Asia and Southern China to alleviate illness. To investigate its immunomodulating properties, composition of polysaccharides and the expression of cytokines/chemokines from *L. rhinocerus* (TM02®) cultivar treated RAW 264.7 were explored. It was revealed, CWE contains linear polysaccharides with 1,4-linkages and rhinopropylcan fraction (HMW & MMW) possesses 1,4-Glcp and 1,6-Glcp backbone and branched chain (1,3,6-Glcp, 1,4,6-Glcp, 1,3,6-Glcp, 1,2,4,6-Glcp). Cytokines profile showed upregulation from CWE (IL-5: 12.078 ± 1.225), HMW (IL-6: 7.297 ± 0.338 ; TIMP-1: 3.358 ± 0.200), MMW (IL-5: 15.412 ± 5.823 ; TIMP-1: 1.747 ± 0.053), and LMW (MIP-2: 3.495 ± 0.416 ; TIMP-1: 7.573 ± 0.088) and possible involvement of NF- κ B and MAPK signaling pathway. Further *in vivo* studies are needed to fully understand the immunomodulatory effects of TM02®.

KEY WORDS: cytokines, chemokines, immunomodulation, *Lignosus rhinocerus*, monosaccharides, glycosidic linkage, medicinal mushrooms

ABBREVIATIONS: 6Ckine, chemokine (C-C motif) ligand 21; CTACK, chemokine (C-C motif) ligand 27; CWE, cold water extract; DMEM, Dulbecco's Modified Eagle Medium; DMSO, dimethyl sulfoxide; Eotaxin-1, C-C motif chemokine 11; FBS, fetal bovine serum; GCSF, granulocyte-colony stimulating factor; GM-CSF, granulocyte-macrophage colony-stimulating factor; HMW, high molecular weight; IFN- γ , interferon gamma; IL, interleukin; LMW, low molecular weight; MCP-1, monocyte chemoattractant protein 1; MCP-5, monocyte chemoattractant protein 5; MIP-1 α , macrophage inflammatory protein 1-alpha; MIP-2, macrophage inflammatory protein 2; MIP-3 β , macrophage inflammatory protein 3-beta; MMW, medium molecular weight; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide; ND, not detected; NO, nitric oxide; nPSPC, novel polysaccharide-protein complexes; sTNFRI, soluble tumor necrosis factor receptor I; TARC, thymus and activation regulated chemokine; TIMP-1, tissue inhibitor of metalloproteinases 1; TNF- α , tumor necrosis factor alpha; VEGF, vascular endothelial growth factor

I. INTRODUCTION

Natural immunomodulators found in medicinal mushrooms have been significantly studied due to their potency and nontoxic characteristics.¹ Immunomodulating compounds from medicinal mushrooms have been

shown to mediate instances of immune deficiency or suppress autoimmune diseases.^{2,3} Researchers have identified the polysaccharides and polysaccharide–protein complexes responsible for their immunomodulating properties. For example, *Ganoderma lucidum* was found to exert antitumor activity on S-180 mouse sarcoma cells via its polysaccharide, β -D-glucan.⁴ Immunomodulatory effects of FVPA1 polysaccharide (extracted from *Flammulina velutipes*) were also uncovered where it was able to stimulate natural killer activity against K562 tumor cells.⁵

To identify the immunomodulating potential of such compounds, the commonly used method is via the induction of cytokine synthesis. Polysaccharides from *Inonotus obliquus* were shown to enhance immunity by increasing mitogenic and comitogenic activities via stimulating the secretion of IFN- γ , IL-4, and TNF- α .⁶ For *Polyporus umbellatus*, it was observed to induce the secretion of IL-1 β and TNF- α , key players in promoting defense against intracellular pathogens.⁷ In terms of immune suppression, *Xylaria nigripes* was shown to inhibit the secretion of inflammatory cytokines TNF- α , IFN- γ , IL-1 β , and IL-6.⁸

In our study, tiger milk medicinal mushroom, *Lignosus rhinocerus* (Cooke) Ryvarden (= *Polyporus rhinocerus*; Polyporaceae, Agaricomycetes), was investigated for its immunomodulatory effects. This mushroom is commonly used by communities in Southeast Asia and Southern China to alleviate various conditions such as cancer, asthma, fever, cough, food poisoning, and gastric ulcer.^{9,10} Procurement of this mushroom is difficult, hence a large-scale cultivation method (solid state fermentation) with controlled environment was introduced by Tan et al.¹¹ to produce optimum quality *L. rhinocerus* termed as TM02[®]. This cultivated mushroom has been shown to have a higher amount of proteins, water-soluble compounds, as well as higher antioxidative effects than the wild type.¹² Based on the increased bioactive compounds, *L. rhinocerus* TM02[®] cultivar was selected for this study.

Besides the differences between the wild type and the cultivated mushroom, other studies on this species have led us to investigate its immunomodulating potential. For example, Wong et al.⁴ reported that the sclerotium of wild type *Polyporus rhinocerus* (a synonym of *L. rhinocerus*) contains various polysaccharides, including water-soluble polysaccharide–protein complexes and alkali-soluble β -glucans that can induce antitumor activity in sarcoma S180-implanted BALB/c mice. It was also revealed to induce cytotoxic effects on various human leukemic cell lines such as HL-60, K562, and THP-1. In a later study, it was determined that its cytotoxic effects on HL-60 were caused by cell cycle arrest at G₁ phase, leading to apoptosis.⁴ Wild type *P. rhinocerus* has also exhibited anti-inflammatory effects *in vitro* by inhibiting lipopolysaccharides-induced TNF- α production in murine macrophage cell RAW 264.7.¹

Despite the strong indication of immunomodulating capabilities of *L. rhinocerus*, there are few studies being done on this specimen especially from the current strain (TM02[®]); instead various articles have been written on notable medicinal mushrooms such as *Trametes versicolor*, *Schizophyllum commune*, and *Ganoderma lucidum*.^{13–15} As a result, we aim to investigate the immunomodulatory properties of *L. rhinocerus* TM02[®] cultivar using murine macrophage cell RAW 264.7. This study was performed by determining the carbohydrate content of *L. rhinocerus* TM02[®] and examining the cytokine profiles of RAW 264.7 cells upon treatment with *L. rhinocerus* TM02[®] cold water extract and their three different molecular weight fractions.

II. MATERIALS AND METHODS

A. Preparation of *L. rhinocerus* TM02[®] Cold Water Extract (CWE)

Pulverized sclerotia powder of *L. rhinocerus* TM02[®] (LiGNO[™] Biotech Sdn. Bhd., Malaysia) was authenticated by DNA fingerprinting.¹⁶ This was mixed with milli-Q water in a ratio of 1 g of powder to 20 ml of water. The mixture was kept at 4°C with continuous stirring for 24 hr. The aqueous mixture was centrifuged (8000g, 30 min) and the supernatant was lyophilized to obtain the CWE.

B. Materials

Analysis grade ammonium acetate, dimethyl sulfoxide (DMSO), and 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) were purchased from Merck Millipore (USA). Dulbecco's Modified Eagle Medium (DMEM; high glucose) media and penicillin–streptomycin mixed solution were purchased from Nacalai Tesque (Japan). Sephadex G-50 fine column was purchased from GE Healthcare (Sweden).

C. Fractionation of *L. rhinocerus* TM02® CWE

The CWE was dissolved in milli-Q water (1 g to 10 ml ratio) and fractionated using the Sephadex G-50 fine column with ammonium acetate. The elution was performed via gravity. The carbohydrate and protein content of each eluate was determined using the phenol-sulphuric acid assay and Bradford assay, respectively.^{17,18} The pooled fractions were grouped into different molecular weights: high molecular weight (HMW > 30 kDa), medium molecular weight (MMW 7-30 kDa), and low molecular weight (LMW < 7 kDa) according to Yap et al.¹⁹ The pooled Sephadex G-50 fractions were subsequently lyophilized.

D. Determination of Carbohydrate Composition and Glycosidic Linkages

The monosaccharide composition of CWE and its Sephadex G-50 fractions was analyzed using alditol acetate derivatives using the Agilent 6890N gas chromatograph equipped with a DB-225 capillary column (15 m × 0.25-mm i.d., 0.25-μm films) and a flame ionization detector. The alditol acetate derivatives were prepared using similar protocol to Pettolino.²⁰ The oven temperature was initially set at 170°C, raised by 2°C/min up to 220°C and held for 15 min. The glycosidic linkages of the carbohydrate samples were analyzed as partially methylated alditol acetate derivatives by an Agilent 6890N gas chromatograph equipped with a HP-5ms GC column (30 m × 0.25 mm, i.d., 0.25 μm films) and a mass spectrometer. The GC conditions used for the analysis were adapted from Zhang et al.²¹ The oven temperature was initially set at 150°C, raised by 3°C/min to 220°C, then raised by 30°C/min to 300°C and held for 10 min.

E. MTT Cytotoxicity Assay

RAW 264.7 cells (ATCC, USA) were cultured in DMEM (high glucose) media with 10% fetal bovine serum (FBS). A single layer of cells was seeded into a 96-well plate (40,000 cells per well) supplemented with 1% penicillin–streptomycin mixed solution. The cells were treated for 24 hr with various concentrations (200 μg/ml, 100 μg/ml, 50 μg/ml, 25 μg/ml, 12.5 μg/ml, and 6.25 μg/ml) from CWE, HMW, MMW, and LMW fractions. After treatment, the spent medium from each well was removed and 0.5 mg/ml of MTT was added into each well and further incubated for 2 hr in the dark. MTT was removed and DMSO was added into each well to dissolve the formazan crystals. The absorbance of each well was read at 570 nm using a Multiskan GO Microplate Spectrophotometer (Thermo Scientific, USA).

F. Determination of Nitric Oxide Production

Nitric oxide (NO) production is determined using Griess reaction.²² NO can oxidize rapidly into nitrate and nitrite when exposed to oxygen. Using Griess reaction, nitrate and nitrite will react with sulfanilamide and *N*-1-naphthylethylenediamine dihydrochloride under acidic condition and produce azo compound

(Chromophore). The procedure used in this study began with preparing a sodium nitrite standard of 0 μM , 1.56 μM , 3.13 μM , 6.25 μM , 12.5 μM , 25 μM , 50 μM , and 100 μM . After that, 50 μl of standard/samples were mixed with 50 μl of Griess reagent for nitrite (Sigma, USA) in a 96-well plate and incubated for 15 min in the dark. The absorbance was read at 550 nm using a Multiskan GO Microplate Spectrophotometer (Thermo Scientific, USA).

G. Detection of Relative Cytokine Profiles

RayBio® Mouse Cytokine Array G2 (RayBiotech Inc., USA) was blocked with 1 $\times$ blocking buffer for 1 hr at room temperature. Samples were added in duplicates followed by incubation at room temperature. After 2 hr, the samples were removed from the slide where the latter was washed three times with wash buffer I and wash buffer II. The wash buffer was completely removed, and biotin-conjugated anticytokines was added into each subarray and incubated for 2 hr at room temperature. The slide was further washed three times with wash buffer I and wash buffer II. Streptavidin-fluor was added into each subarray with an incubation time of 2 hr at room temperature in the dark. The Streptavidin-fluor was removed, and the slide was washed three times with wash buffer I and wash buffer II. The slide was then air dried and scanned with an excitation frequency of 532 nm using Cy3-compatible (green) InnoScan laser scanner (Innopsys, Singapore).

H. Determination of Cytokines

A total of 100 μl of sample/standard was added into each well of a 96-well plate (clear flat bottom), and incubated for 2.5 hr at 37°C. The sample/standard was removed, and biotin-antibody was added into each well, incubated for 1 hr at 37°C. Then, each well was washed three times before adding HRP-avidin. After another incubation for 1 hr at 37°C, each well was washed five times before substrate was added. The plate was incubated for 30 min at 37°C. Finally, stop solution was added, and the plate was read at 450 nm using Multiskan GO Microplate Spectrophotometer (Thermo Scientific, USA).

I. Statistical Analysis

Cytokine array and ELISA were performed in duplicates while MTT assay was conducted in triplicates. Results are expressed as mean $\pm$ SD. Differences between each group were analyzed using Statistical Package for Social Sciences (SPSS) statistic version 22 (IBM®, USA) under ANOVA, LSD T-test, and Tukey HSD test with a $p < 0.05$ considered as significant.

III. RESULTS

A. Monosaccharide Composition and Methylation Linkage Analysis

Results of the GC analysis revealed that the carbohydrates in the mushroom extracts were predominantly glucose, with a very small amount of galactose and mannose as well as some pentoses such as fucose, xylose, and arabinose (Table 1). Therefore, the polysaccharides in the mushroom extracts are presumably glucans. From the results of the methylation linkage analysis (Table 2), there were no glycosidic linkages detected in the LMW fraction, implying that this fraction contained mainly noncarbohydrate components. The glucans found in CWE are linear polysaccharides with 1,4-linkage while the MMW and HMW contained highly branched glucans with mixed linkages.

TABLE 1: Monosaccharide composition of CWE and fractions

Sample	Total Carbohydrate %*	Fuc	Ara	Xyl	Man	Gal	Glc
TM02®							
1. CWE	61.54	ND	0.73	0.70	1.67	1.91	94.99
2. LMW	38.61	3.82	0.65	0.65	2.31	ND	92.56
3. MMW	52.79	2.76	0.52	0.50	1.70	2.30	92.21
4. HMW	76.85	1.63	0.56	0.68	1.23	1.36	94.53

*Data are expressed in percentage (%) as average value of duplicate measurements in percentage.

TABLE 2: Glycosidic linkages of CWE and fractions

Glycosidic linkage	T-Glcp	T-Galp	1,4-Glcp	1,6-Glcp	1,4,6-Glcp	1,3,6-Glcp	1,2,4,6-Glcp
TM02®							
1.CWE	19.89	14.21	65.90	ND	ND	ND	ND
2.LMW	ND	ND	ND	ND	ND	ND	ND
3.MMW	13.68	9.56	49.13	4.26	8.17	3.83	11.37
4.HMW	19.26	13.87	52.02	10.35	ND	3.90	ND

Data are expressed in percentage (%) as average value of duplicate measurements in percentage.

B. Total Protein Content of CWE and Its Fractions

The total protein content of CWE and its fractions were analyzed via the Bradford assay (Table 3). The total protein content in all *L. rhinocerus* TM02® fractions were less than 10% of the dry weight with HMW having the highest amount.

C. RAW 264.7 Macrophage Cell Activation upon *L. rhinocerus* TM02® Treatment

To evaluate the immunomodulating activity of CWE and its fractions, prior activation of RAW 264.7 cells was required. RAW 264.7 cells were treated for 24 hr with CWE, HMW, MMW, and LMW of different concentrations (200 µg/ml, 100 µg/ml, 50 µg/ml, 25 µg/ml, 12.5 µg/ml, and 6.25 µg/ml). Cell proliferation was determined via MTT assay. LMW does not affect the proliferation of cells while CWE, HMW, and MMW exhibited significant effects in a dose-dependent manner. At a dose of 12.5 µg/mL, CWE (cell viability: 75%) showed the highest inhibition of proliferation followed by HMW (cell viability: 77%) and MMW (cell viability: 95%). This dosage was then used for treatment of cells to be used for subsequent analyses (Sections D, E, F).

TABLE 3: Protein content of CWE and fractions

	CWE	LMW	MMW	HMW
Protein content (%)	2.02	0.40	1.82	6.01

Data are expressed as average value of duplicate measurements.

D. Determination of NO Production in RAW 264.7 Cells Treated with CWE and Its Fractions

No detectable level of NO was present in RAW 264.7 cells after treatment with CWE and its fractions (data not shown). Similar observation was noted in the negative control where the RAW 264.7 cells were not treated with any of the extract and fractions.

E. Cytokine Array Profile of RAW 264.7 Cells Treated with CWE and Its Fractions

Table 4 shows the cytokine/chemokine fold changes in RAW 264.7 cells upon treatment with CWE and its various fractions. CWE treatment resulted in a significant decrease in both IL-4 and IL-5 levels, and a significant increase in IL-6, IL-9, and MCP-5 levels. HMW treatment produced significant increases in IL-6, TARC, and TIMP-1 levels; MMW showed significant increase in IL-5, IL-10, TIMP-1 levels; and LMW treatment showed significant increase in IL-5, IL-10, IL12 p40p70, MCP-5, and MIP-2 levels.

F. Verification of Selected Cytokines and Chemokines

ELISA assay for selected cytokines/chemokines was used to verify the results in Table 4 as the antibody array only provides an estimate quantitative reading of cytokine/chemokine levels. IL-5, IL-6, MIP-2, and TIMP-1 were selected for verification due to their significant results. We confirmed that IL5 and TIMP-1 are upregulated in RAW 264.7 cells when treated with CWE, especially for IL5 with a significant fold change of 12.078 ± 1.225 (Table 5). Upregulation of IL-5, IL-6, MIP-2, and TIMP-1 were validated in HMW treatment. As for MMW treatment, a significant upregulation of IL-5 (15.412 ± 5.823) was evident, with minor upregulation of IL-6 and TIMP-1. LMW treatment only resulted in the upregulation of MIP-2, IL-6, and TIMP-1.

Compared with the significant decrease of IL-5 levels (fold change of 0.40 ± 0.01) in the cytokine array, the ELISA assay revealed a significant increase (fold change of 12.078 ± 1.225). In addition, the cytokine array indicated an upregulation of IL-6, MIP-2, and TIMP-1 expression upon CWE treatment but the ELISA assay later revealed that it does not significantly affect IL-6 and MIP-2 secretion. As for both HMW and MMW treatment, the fold changes observed between the cytokine array and ELISA assay were comparable, albeit higher in the latter. This suggests that the HMW and MMW fractions may have more immunomodulating capabilities than what was previously revealed.

IV. DISCUSSION

In present times, natural immunomodulating agents are highly sought after to enhance our immune system against pathogens and cancer and to manage autoimmune diseases. In our study, *L. rhinocerus* TM02[®] was extracted using cold water extraction to isolate as much bioactive compounds as possible since thermolabile compounds tend to degrade during hot water extraction.²³ Prior study on this cultivar has demonstrated CWE to have higher bioactivity (antiproliferative and antioxidative) properties compared with hot water extract and methanol extract.¹² Not only that, water extraction further allows the isolation of water-soluble polysaccharides that primarily play a role in immunomodulation and antitumor activities.²³ The current *in vitro* study on *L. rhinocerus* TM02[®] showed regulation of cytokines and chemokines. RAW 264.7 murine macrophage cells were used for the *in vitro* study due to their involvement in both innate and adaptive immunity as well as regulation of tissue development, repair, remodeling, and homeostasis via secretion of various cytokines, chemokines, and growth factors.¹

Previous study of hot water crude extract of a wild type *L. rhinocerus* (= *P. rhinoceros*) revealed lower carbohydrates (21.3%) but higher protein (44.2%) content than the current study of CWE (61.54% and

TABLE 4: Cytokine/chemokine secreted by RAW 264.7 upon treatment with CWE, HMW, MMW and LMW fractions of *Lignosus rhinocerus* TM02®

Cytokine/ chemokine	Fold changes			
	CWE	HMW	MMW	LMW
Pro-inflammatory				
6Ckine	1.58 ± 0.03	1.09 ± 0.04	1.31 ± 0.18	1.36 ± 0.10
CTACK	8.28 ± 8.40	1.02 ± 0.05	1.29 ± 0.45	1.16 ± 0.01
GM-CSF	1.15 ± 0.42	1.12 ± 0.01	1.17 ± 0.20	1.21 ± 0.04
IL-2	1.98 ± 1.64	1.14 ± 0.02	1.08 ± 0.21	0.81 ± 0.21
IL-6	2.38 ± 0.07*	2.61 ± 0.37*	1.38 ± 0.32	1.59 ± 0.23
IL-12 p40p70	0.69 ± 0.03	1.24 ± 0.01	1.33 ± 0.37	1.57 ± 0.18*
IL-12p70	0.86 ± 0.09	1.02 ± 0.09	1.07 ± 0.21	1.05 ± 0.08
IL-17	1.06 ± 0.24	1.09 ± 0.02	1.04 ± 0.21	1.03 ± 0.04
IFN-γ	1.52 ± 0.50	1.13 ± 0.08	1.11 ± 0.18	1.03 ± 0.03
MCP-1	1.64 ± 0.31	0.77 ± 0.08	0.88 ± 0.13	0.81 ± 0.08
MCP-5	4.74 ± 0.64*	1.33 ± 0.01	1.12 ± 0.15	1.83 ± 0.90*
MIP-1α	0.73 ± 0.18	0.66 ± 0.06	0.76 ± 0.08	0.88 ± 0.01
MIP-2	1.71 ± 0.10	1.00 ± 0.05	1.14 ± 0.21	2.58 ± 0.09*
TNF-α	0.83 ± 0.20	1.14 ± 0.11	1.29 ± 0.28	1.43 ± 0.17
Thrombopoietin	9.94 ± 9.98	1.11 ± 0.19	1.41 ± 0.18	1.26 ± 0.30
VEGF	N/A	1.27 ± 0.00	1.82 ± 0.45	2.25 ± 1.29
Anti-inflammatory				
GCSF	1.44 ± 2.03	N/A	N/A	N/A
IL-3	0.98 ± 0.10	1.22 ± 0.08	1.34 ± 0.08	1.33 ± 0.21
IL-4	0.60 ± 0.03 [#]	1.18 ± 0.06	1.31 ± 0.30	1.30 ± 0.14
IL-5	0.40 ± 0.01 [#]	1.77 ± 0.04	2.28 ± 0.63*	2.27 ± 0.05*
IL-9	2.12 ± 0.10*	1.23 ± 0.01	1.29 ± 0.35	1.28 ± 0.01
IL-10	2.46 ± 0.44	1.32 ± 0.01	1.53 ± 0.37*	1.61 ± 0.11*
sTNFRI	1.41 ± 0.22	0.76 ± 0.07	0.86 ± 0.16	0.73 ± 0.04
TARC	1.97 ± 0.77	2.03 ± 0.15*	1.48 ± 0.41	1.33 ± 0.27
TIMP-1	3.46 ± 3.01	1.90 ± 0.47*	1.81 ± 0.08*	1.05 ± 0.16

Data are expressed as average value of duplicate measurements representing fold changes of cytokine/chemokine relative to negative control. Data in the column are analyzed under *post hoc* LSD test to determine significant increase and decrease relative to negative control. * $P < 0.05$ vs. negative control (significant increase). [#] $P < 0.05$ vs. negative control (significant decrease).

2.02%, respectively).¹ This may be due to differences in the extraction temperature and chemical composition of the two different strains. Carbohydrate composition of water extract fractions from wild type *L. rhinocerus* was predominantly glucose (91.2%) and mannose (8.8%) while that from *L. rhinocerus* TM02® cultivar was predominantly glucose (92–95%) with small amounts of mannose (1–2%), galactose (1–2%), and very small amounts of fucose, xylose, and arabinose.¹

Compared with the glycosidic linkages found in wild type *L. rhinocerus* (comprising similar molar ratios of mixed linkages¹), the *L. rhinocerus* TM02® MMW and HMW fractions (also the same fraction containing

TABLE 5: Fold change of selected cytokine/chemokine secreted by treated RAW 264.7 cell

Extract	IL-5	IL-6	MIP-2	TIMP-1
CWE	12.078 ± 1.225*	0.595 ± 0.000	0.686 ± 0.167	1.201 ± 0.038
HMW	2.471 ± 0.832	7.297 ± 0.338*	1.173 ± 0.014	3.358 ± 0.200*
MMW	15.412 ± 5.823*	1.405 ± 0.000	0.883 ± 0.568	1.747 ± 0.053*
LMW	ND	> 204	3.495 ± 0.416*	7.573 ± 0.088*

Data are expressed as average value of duplicate measurements representing fold change in cytokine/chemokine relative to negative control. Data in the column was analyzed under *post hoc* LSD test to determine significant changes relative to negative control. * $P < 0.05$ vs. negative control (significant increase).

novel bioactive proteins of *L. rhinocerus* TM02^{®24,25}) contain abundant amounts of highly branched glucans dominated by 1,4-linkage. This suggests that the polysaccharide backbones of both fractions are made up of 1,4-Glcp as well as 1,6-Glcp. Our results have revealed the presence of highly branched polysaccharides in the MMW fraction, consisting of 1,4,6-Glcp, 1,3,6-Glcp, and 1,2,4,6-Glcp, whereas the HMW fraction contains 1,3,6-Glcp only. A similar backbone was found in GCP-2, a water soluble β -glucan derived from *Ganoderma sinense* that is capable of inducing B-cell proliferation, secretion of nitric oxide from macrophages, and secretion of inflammatory cytokines from human peripheral blood mononuclear cells.²⁶ Although this may suggest that the polysaccharides found in MMW and HMW possesses similar immunomodulatory capabilities as GCP-2, other features such as the degree of polymerization, degree of branching, and the length of its various side chains may affect the function of these polysaccharides.^{26,27} The MMW and HMW are henceforth known as the rhinopropykan fraction containing novel polysaccharide–protein complexes (nPSPC).

With regard to the cytokines examined in this study, IL-5 is involved in growth, differentiation, and activation of eosinophils.²⁸ Eosinophils help enhance the immune system against microorganisms and parasitic helminths and even contribute to the pathogenesis of asthma, as demonstrated in a study where IL-5⁻mice treated with recombinant vaccinia viruses for IL-5 demonstrated restored pulmonary inflammation and airway hyperreactivity.²⁹ This resulted in numerous studies for targeting IL-5 for inhibiting asthmatic response.^{28,30} There are indications of eosinophils possessing antitumor abilities as they have been shown to have cytotoxic activity against tumor cells.³¹ In our study, CWE, HMW, and MMW were able to stimulate upregulation of IL-5 secretion. Further confirmation *in vivo* is needed to observe whether *L. rhinocerus* TM02[®] cultivar may promote pro-inflammatory responses and other associated synergistic effects.

IL-6 was shown to be stimulated by all treatments. Interestingly, in the ELISA study, LMW showed a very high figure. We postulate that the high value may in part be due to activation by secondary metabolites that are abundantly found in the LMW leading to overexpression of IL-6 as these metabolites are also known to immunomodulate. IL-6 is considered as a pro-inflammatory cytokine that can induce B cell differentiation as well as differentiation of T cells into T_H17. T_H17 is responsible for secreting IL-17, which helps regulate bacterial and fungal infections.³²

IL-6 is also involved in rheumatoid arthritis (RA), where osteoclasts are differentiated and activated, causing bone resorption leading to osteoporosis.³³ It also enhances VEGF production and triggers angiogenesis; increase in vascular permeability contributes to inflammatory lesions observed in RA. Although there are many studies indicating IL-6 as a pro-inflammatory agent, other studies have shown that it possesses anti-inflammatory properties depending on the physiological systems involved. A study by Tilg et al.³⁴ has shown that inducing the secretion of adrenocorticotrophic hormone in the central nervous system stimulates production of glucocorticoids that suppress pro-inflammatory cytokine (such as IL-1 and TNF) production. An increased in IL-6 secretion may promote tumor growth. For example, BALB/c mice implanted with IL-6 inducer caused the formation of plasma cell tumors.³⁵

Macrophage inflammatory protein 2 (MIP-2) belongs to the CXC chemokine family where it has strong chemotactic activity for neutrophils. A study on BCG-primed mice treated with anti-MIP-2 revealed significantly lower neutrophil accumulation.³⁶ An increase of MIP-2 expression aids the immune system to attract neutrophils to counteract infection. Another study on MIP-2 has demonstrated mitogenic stimulation in epithelial cells, indicative of wound healing.³⁷ In a study on extrahepatic colorectal metastasis, results revealed a strong indication of tumor proliferation, growth, and invasiveness where MIP-2 exposure increased the expression of CXCR2 receptor, promoting tumor growth.³⁸ This latter study indicates the possibility of MIP-2 in stimulating tumor-promoting capabilities. In our results, the CWE, HMW, and MMW treatment showed no significant changes in MIP-2 expression compared with the significant increase observed in LMW treatment.

Regarding the modulatory secretion of IL-5, IL-6, and MIP-2 from macrophages after treatment with HMW and MMW, there is a possibility that these cytokines are linked to the IL-17 signaling pathway. IL-6-induced secretion of IL-17 signals macrophages to partake in NF- κ B and MAPK signaling pathway.^{32,39} The NF- κ B signaling pathway can induce the transcription T_H2 cytokines such as IL-4, IL-5, and IL-13 as well as CXC chemokines (such as MIP-2) and inflammatory cytokines (such as IL-6) to stimulate T_H2 response and neutrophil recruitment, respectively. Like NF- κ B signaling pathway, MAPK signaling can also induce transcription of CXC chemokines and inflammatory cytokines to enhance immune responses against pathogens.

Our current study revealed an increase in expression of TIMP-1. TIMP-1 is responsible for the inhibition of matrix metalloproteinases (MMPs).⁴⁰ MMPs' role is to degrade the extracellular matrix surrounding tissues to allow tissue remodeling and mitogenic activity of cells for wound healing. In addition, MMPs have been shown to promote tumor cell propagation and angiogenesis, leading to RA complications.^{41,42} The inhibitive properties of TIMP-1 prevented further metastasis of tumor cells and inhibited the progression of RA. In an *in vivo* study using mice, inhibition of TIMP-1 with antibody resulted in a twofold increase in invasion of blood vessels relative to mice receiving IgG.⁴² TIMP-1 can also promote growth. A study has shown that TIMP-1 possesses sequence homology with human granulocyte-macrophage colony stimulating factor (GM-CSF), a type of growth factor.⁴³ The same study also showed that human gingival fibroblasts (Gin-I) and Burkitt lymphoma cells (Raji) cultured in serum-free media do not grow in basal media DMEM and RPMI 1640 but show significant proliferation in media supplemented with TIMP-1.⁴³ Studies on rabbit corneal epithelial cells and human keratinocytes have also revealed increase in cell growth when cell culture medium was supplemented with TIMP-1.^{44,45} All fractions in our study were shown to cause a significant increase in TIMP-1 levels, especially HMW (3.358 ± 0.200) and LMW (7.573 ± 0.088). These fractions may possess more immunomodulatory capacity than CWE.

In the current study, details on the exact components responsible for changes in specific cytokines/chemokines secretion are still lacking. The specific effect of upregulating IL-5, IL6, MIP-2, and TIMP-1 expression on the biological system would need to be further explored, as well as the causative agent in the cultivar that causes these upregulations. This necessitates further investigation to understand the precise immunomodulatory function of *L. rhinocerus* TM02[®] cultivar. On top of the cytokine responses by RAW 264.7, NO production from macrophage cells was also studied because NO secretion can contribute to antimicrobial and antitumor activities. Based on the results we found that none of the fractions nor CWE can induce NO production. This might prove beneficial as NO production can contribute to oxidative stress as well as act as an immunosuppressor.^{46–48}

V. CONCLUSIONS

The structural composition of polysaccharides found in CWE were linear polysaccharides with 1,4-linkages whereas polysaccharides from rhinoproplycan fraction contain backbones constructed

with 1,4-Glcp and 1,6-Glcp branched with (HMW: 1,3,6-Glcp) and (MMW: 1,4,6-Glcp, 1,3,6-Glcp and 1,2,4,6-Glcp). These β -glucans can enhance the immune system. The rhinopropyran fraction of *L. rhinocerus* TM02[®] cultivar seem to immunomodulate RAW 264.7 by stimulating IL-5, IL-6, MIP-2, and TIMP-1 expression, with a possible involvement in the NF- κ B and MAPK signaling pathway. The extract and fractions were capable in enhancing pro-inflammatory reactions, suggesting that *L. rhinocerus* TM02[®] cultivar possesses immunomodulating activities. In addition, all treatment does not affect NO production from RAW264.7 that might contribute to oxidative stress. Further *in vivo* studies are needed to fully understand the effects of *L. rhinocerus* TM02[®] on the immune system.

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